Snehal Kumar Ketala

+1 (214)735-2610 | snehalkumarketala@gmail.com | LinkedIn | Github

Professional Summary:

- Lead Data Engineer with **5+ years of experience** designing and optimizing enterprise-scale data platforms on **AWS (S3, Glue, Redshift, Athena, EMR, Lambda)** to deliver secure, scalable, analytics-ready data products across domains.
- Led and mentored engineering teams to build robust **ETL/ELT frameworks using AWS Glue, Step Functions, and Lambda**, automating ingestion and improving reliability by 70% while aligning with governance standards.
- Architected cloud-native **lakehouse ecosystems on S3, Redshift Spectrum, and Apache Iceberg**, improving query performance by 45% and reducing storage costs through compression and partition optimization.
- Developed real-time streaming solutions using **Kinesis, Kafka (MSK), and EMR Spark**, processing millions of messages daily with sub-second latency for fraud detection and operational insights.
- Implemented **CI/CD and Infrastructure-as-Code** automation via **Terraform, CloudFormation, and CodePipeline**, ensuring reproducible environments and zero-downtime deployments for enterprise pipelines.
- Optimized large-scale **Spark on EMR** workloads through Adaptive Query Execution, broadcast joins, and partition pruning, reducing runtime by 40% and AWS EC2 compute costs by 25%.
- Enhanced compliance and governance with **IAM, Lake Formation, KMS encryption**, and RBAC, ensuring **HIPAA and PCI-DSS** alignment and securing sensitive datasets across AWS environments.
- Engineered end-to-end **Airflow orchestration pipelines** integrated with **AWS Glue and Redshift**, achieving 99.9% availability and consistent data synchronization across critical workloads.
- Established automated **data quality frameworks using Great Expectations and Glue Data Catalog**, improving schema validation accuracy by 80% and preventing production anomalies.
- Built dimensional and analytical models within **Redshift and Snowflake**, enabling faster ad-hoc reporting and cutting dashboard latency by 50% through efficient query design and optimization.
- Deployed scalable microservices using **ECS, EKS, and API Gateway** integrated with **CloudWatch and OpenTelemetry**, achieving p95 latency under 200 ms and full observability.
- Partnered with architects and DevOps to perform **architecture reviews, cost optimization, and AWS resource tuning**, ensuring alignment with enterprise data strategy and KPIs.
- **AWS Certified Data Analytics – Specialty** professional skilled in **ETL, Terraform, Airflow, Redshift, and Lake Formation**, driving data modernization and business intelligence initiatives.

Technical Skills:

Programming & Scripting: Python, PySpark, Scala, Java, SQL, Shell Scripting

AWS Cloud & Data Services: S3, Glue, Redshift, Athena, EMR, Lambda, Lake Formation, Kinesis, MSK, Step Functions, ECS, EKS, CloudFormation, CodePipeline, CloudWatch, IAM, KMS

Data Engineering & Lakehouse: ETL/ELT Frameworks, Data Lakes, Apache Iceberg, Delta Lake, Dimensional Modeling, Redshift Spectrum, Data Catalog, Data Governance

Workflow Orchestration: Apache Airflow, AWS Step Functions, Amazon Managed Workflows (MWAA)

Infrastructure as Code & CI/CD: Terraform, CloudFormation, Jenkins, GitHub Actions, CodeBuild, CodeDeploy

Data Modeling & Databases: Redshift, Snowflake, PostgreSQL, MySQL, Oracle, DynamoDB, Cassandra, MongoDB

Data Streaming & Messaging: Kinesis Data Streams, Kafka (MSK), Firehose, SNS, SQS

Data Quality & Observability: Great Expectations, OpenMetadata, OpenTelemetry, CloudWatch Metrics, Log Insights

Machine Learning & Analytics: SageMaker, EMR Spark MLlib, Athena SQL Analytics, Python ML Frameworks (scikit-learn, XGBoost)

Security & Compliance: IAM Roles & Policies, Lake Formation RBAC, KMS Encryption, VPC Endpoints, HIPAA/PCI-DSS Alignment

Development & Version Control: Git, Bitbucket, Docker, Kubernetes, VS Code, JupyterLab
BI & Visualization: Power BI, Tableau, Amazon QuickSight, Looker

Professional Experience:

***Data and Generative AI Engineer* | Toyota Motor Corporation | Texas, USA** **Aug 2024 – Present**

Project: LLM-Powered Supplier Quality Assistant – Enterprise Data & GenAI Platform on **AWS (S3, Glue, Redshift, EMR, Lambda, Lake Formation, ECS, CloudFormation)**

- Led modernization of Toyota's supplier quality ecosystem using **AWS and GPT-4 RAG pipelines**, automating query insights and improving data retrieval speed by 70% across global teams.
- Built scalable **ETL/ELT pipelines using AWS Glue, Step Functions, and Lambda**, automating supplier data ingestion from 50K+ PDFs and increasing processing throughput by 60%.
- Built a **data lakehouse on S3 and Redshift Spectrum** leveraging **Apache Iceberg**, optimizing analytics performance by 45% and reducing data duplication by 30%.
- Developed **FastAPI microservices on ECS with KMS encryption and IAM policies**, ensuring secure and scalable GenAI API endpoints with p95 latency under 200 ms.
- Implemented **RAG workflows using LangChain, OpenAI GPT-4, and Redshift vector store**, boosting retrieval precision by 60% and reducing manual lookup efforts significantly.
- Automated orchestration with **Airflow and EMR Spark**, enabling end-to-end pipeline execution, retraining, and validation at 99.9% reliability and near-zero failure rate.
- Enhanced governance using **Lake Formation, IAM roles, and KMS keys**, achieving **HIPAA and PCI-DSS compliance** for supplier and engineering data across AWS.
- Established monitoring via **CloudWatch, OpenTelemetry, and Glue Data Catalog**, improving observability, lineage tracking, and reducing incident response time by 50%.
- Implemented **CI/CD pipelines using Terraform and CloudFormation**, standardizing deployments and cutting environment provisioning time from days to hours.
- Collaborated with AI architects to review **AWS resource utilization and cost metrics**, optimizing EMR and ECS configurations, saving 25% on compute expenses.
- Designed a **LangChain-powered prompt orchestration engine** for GPT-4 and internal APIs, reducing response inconsistencies by 40% across GenAI query flows.
- Delivered production-ready GenAI platform integrating **AWS Lake Formation, Redshift, and GPT-4**, streamlining supplier analysis workflows and enhancing decision accuracy.

***Data Engineer* | Discover Financial Services | Texas, USA** **Jan 2024 – Jul 2024**

Project: Credit Default Prediction – End-to-End ML Pipeline and Analytics Platform on **AWS (S3, Glue, Redshift, SageMaker, EMR, Lambda, Airflow, Terraform)**

- Engineered scalable **ETL pipelines with AWS Glue and Step Functions** to process 10TB+ transactional data from S3 to Redshift, improving data refresh latency by 65% and enabling real-time portfolio insights.
- Designed and deployed a **risk analytics pipeline on EMR Spark, Lambda, and Redshift**, automating model retraining workflows and reducing manual engineering effort by 30%.
- Engineered 50+ behavioral and financial features using **PySpark and SQL**, enhancing predictive accuracy and improving model ROC-AUC by 15% for credit risk identification.
- Implemented **data validation and anomaly detection using Great Expectations**, reducing schema drift and improving production data consistency by 70%.
- Built secure inference APIs with **Lambda, API Gateway, and SageMaker endpoints**, achieving p95 latency under 200 ms and 99.9% uptime for credit risk predictions.
- Automated orchestration using **Airflow and Terraform**, integrating data extraction, transformation, and model execution while ensuring complete audit traceability.

- Integrated **SHAP and SageMaker Clarify** for explainability, improving model transparency and regulatory compliance for business-critical credit models.
- Tuned **Spark jobs on EMR** via AQE, partition optimization, and broadcast joins, reducing compute cost by 25% and runtime by 40% across ML feature pipelines.
- Deployed **infrastructure-as-code using Terraform and CloudFormation**, standardizing provisioning of Glue, S3, and Redshift environments and reducing setup time by 80%.
- Developed analytical marts on **Redshift Spectrum and Athena**, enabling high-performance reporting and improving query efficiency by 50% for finance teams.
- Built observability stack using **CloudWatch, OpenTelemetry, and SNS notifications**, enabling proactive anomaly alerts and cutting incident recovery time by 50%.
- Collaborated with DevOps and Data Science teams to implement **CI/CD, governance, and lineage frameworks**, ensuring secure, auditable, and scalable AWS pipelines.

***Data Engineer* | Medline | Hyderabad, India** **Mar 2020 – Aug 2023**

Project: Healthcare Data Migration and Analytics Modernization on **AWS (S3, Glue, Redshift, EMR, Lambda, Lake Formation, CloudWatch, Terraform)**

- Migrated 10+ years of healthcare EHR and claims data to **AWS S3 and Redshift**, improving scalability by 40%, optimizing analytics performance, and reducing legacy system dependency by 60%.
- Designed and automated **ETL pipelines using AWS Glue, Step Functions, and Lambda**, handling incremental loads with watermarking, achieving 99.9% reliability and full audit traceability.
- Implemented **data lakehouse architecture using S3, Redshift Spectrum, and Apache Iceberg**, enabling ACID compliance, schema evolution, and faster analytical queries.
- Developed **PySpark pipelines on EMR** with partition pruning, broadcast joins, and AQE, cutting ETL runtime by 50% and lowering compute cost by 30% across production workloads.
- Integrated **Lake Formation and IAM-based RBAC** for fine-grained access control, ensuring **HIPAA compliance** and secure handling of PHI across all healthcare datasets.
- Established automated **data validation and reconciliation** using Great Expectations, Glue Catalog, and SNS alerts, reducing data integrity issues and improving pipeline trust by 80%.
- Built near real-time analytics dashboards in **Amazon QuickSight and Power BI** on Redshift, cutting reporting time by 60% and enhancing decision-making for clinical teams.
- Deployed **Airflow DAGs orchestrating Glue, Lambda, and EMR jobs**, enabling seamless ingestion, transformation, and validation workflows with complete observability.
- Integrated **CloudWatch and OpenTelemetry** for monitoring, alerting, and log aggregation, improving mean-time-to-recovery (MTTR) and system uptime by 45%.
- Provisioned infrastructure using **Terraform and CloudFormation**, ensuring reproducible deployments and consistent configurations across staging and production.
- Partnered with governance teams to design **PHI encryption and KMS key rotation** strategies, ensuring secure data exchange and full regulatory compliance alignment.
- Delivered a HIPAA-aligned **AWS data modernization framework** supporting analytics, compliance, and data governance goals while accelerating insights delivery by 50%.

Certifications:

- **AWS Certified Machine Learning Engineer – Associate**
- **AWS Generative AI Applications Professional Certificate**
- **AWS Certified Data Analytics**
- **Databricks Certified: Data Engineer Associate**
- **Databricks Certified: Generative AI Engineer Associate**
- **Databricks Certified: Machine Learning Associate**
- **Microsoft Certified: Azure Data Scientist (DP-100)**
- **Microsoft Certified: Azure Data Engineer (DP-203)**
- **Microsoft Certified: Azure AI Fundamentals**

Projects:

Real-Time Fraud Streaming | AWS EMR, Kinesis, MSK, Airflow, Snowflake, Terraform

- Engineered a **real-time fraud detection pipeline** using **PySpark on AWS EMR**, **Kinesis**, and **MSK**, enabling **sub-second scoring** with **99.9% uptime** for financial transactions.
- Orchestrated scalable **Apache Airflow DAGs** to manage streaming ingestion, reconciliation, and **Snowflake writes**, ensuring strict **SLA compliance** across all pipelines.
- Implemented **Terraform-based autoscaling** for **EMR clusters**, reducing compute spend by **35%** while maintaining **p50/p95 latency targets under 500 ms** consistently.
- Integrated **AWS CloudWatch** and **OpenTelemetry** for real-time metrics, anomaly detection, and error tracing, improving **system observability by 40%** across workloads.
- Delivered a **production-grade fraud prevention platform** powering **risk analytics dashboards**, improving **operational responsiveness by 60%** organization-wide.
- Optimized **Kafka topic partitions** and **Spark structured streaming checkpoints**, achieving **35% higher throughput** and enhanced fault tolerance under high load conditions.
- Automated data validation and reconciliation workflows with **Great Expectations** and **Airflow sensors**, ensuring **zero message loss** and **100% event accuracy** across streams.

Fuzzy Logic Migraine Prediction Using LSTM and Multilayer Perceptron | Python, TensorFlow, Fuzzy Logic

- Developed a hybrid **LSTM and MLP deep learning model** with integrated **fuzzy logic components** in **Python**, improving migraine prediction accuracy and interpretability.
- Achieved **93% accuracy** with **LSTM** and **90% with MLP**, proving recurrent architectures outperform dense models on temporal healthcare data for early migraine detection.
- Designed fuzzy membership and rule-based inference logic to manage uncertainty, enhancing decision transparency and increasing clinician confidence in predictions.
- Implemented end-to-end **TensorFlow pipelines** for model training, validation, and tuning, reducing convergence time and improving model reproducibility by **25%**.
- Integrated **scikit-fuzzy** and **NumPy** modules to build interpretable fuzzy systems, allowing granular thresholding and contextual mapping of patient-specific signals.
- Applied **time-series preprocessing** using **pandas** and **Signal Smoothing** to handle noisy input features, improving data quality and predictive performance.
- Optimized model deployment using **ONNX** and **TensorFlow Lite**, reducing model inference time by **35%** for real-time clinical prediction readiness.
- Delivered an interpretable **AI-driven migraine prediction platform**, supporting healthcare analytics research and early intervention strategies for patient well-being.

Education:

The University of Texas at Dallas | Richardson, TX
Master of Science in Computer and Information Sciences (Major: Data Science)
SRM Institute of Science and Technology | Chennai, India
Bachelor of Technology in Computer Science and Engineering (Major: Big Data Analytics)